4 It would take 120 minutes to fill a swimming pool using water from 5 taps.

(a) How many minutes will it take to fill the pool if only 3 of the taps are used?

5 taps × 120 min = 600 tap-minutes

1 mark
Same volume: taps × time is constant

3 taps: 600 ÷ 3 = 200 minutes

1 mark
Fewer taps need more time

200 minutes

(2)

(b) State one assumption you made in working out your answer to part (a).

All taps deliver water at the same constant rate.

1 mark
State the equal-flow assumption

(1)

(Total for Question 4 is 3 marks)

5 A plane travels at a speed of 213 miles per hour.

(a) Work out an estimate for the number of seconds the plane takes to travel 1 mile.

seconds

(3)

(b) Is your answer to part (a) an underestimate or an overestimate?
Give a reason for your answer.

(1)

(Total for Question 5 is 4 marks)

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